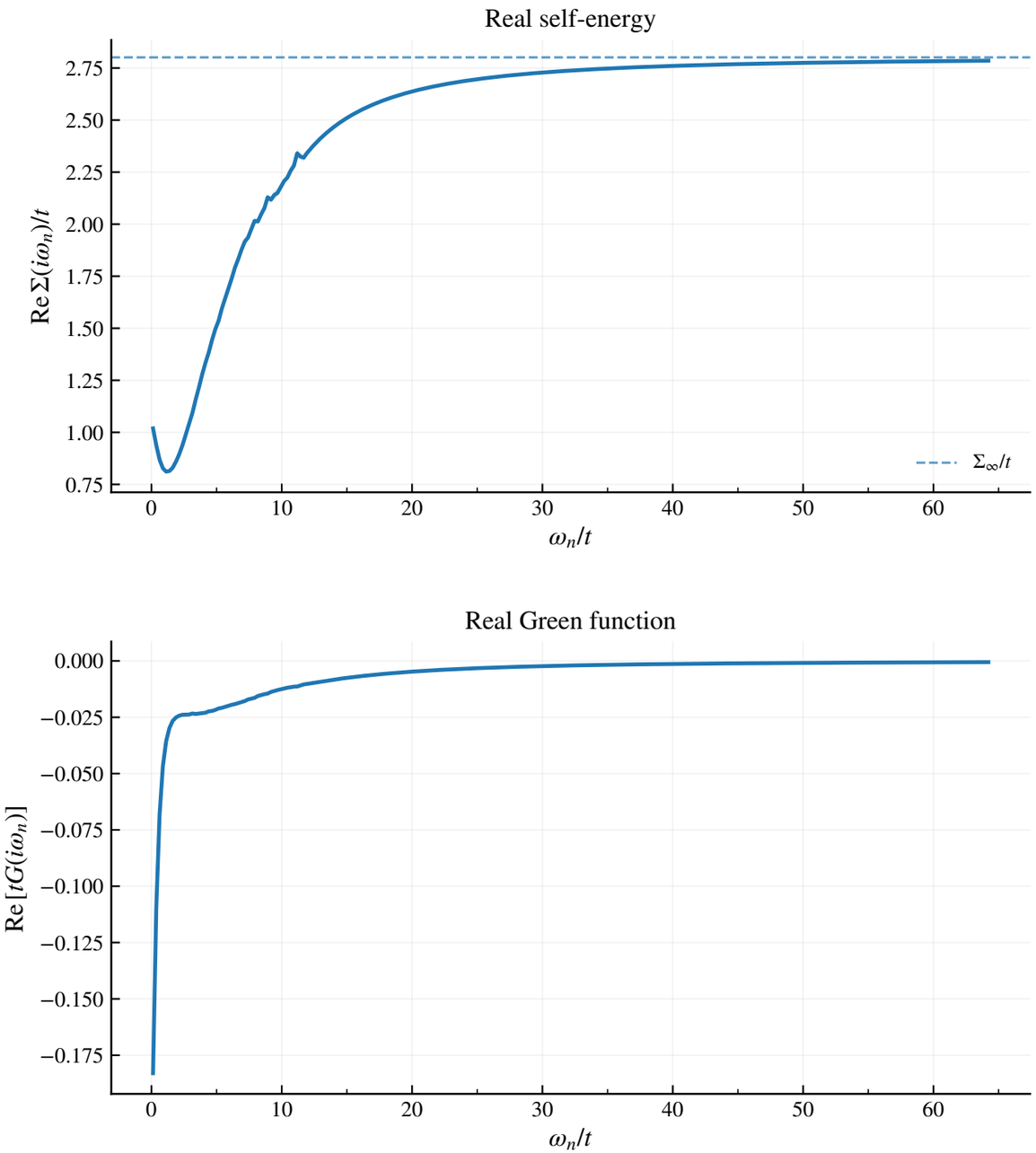
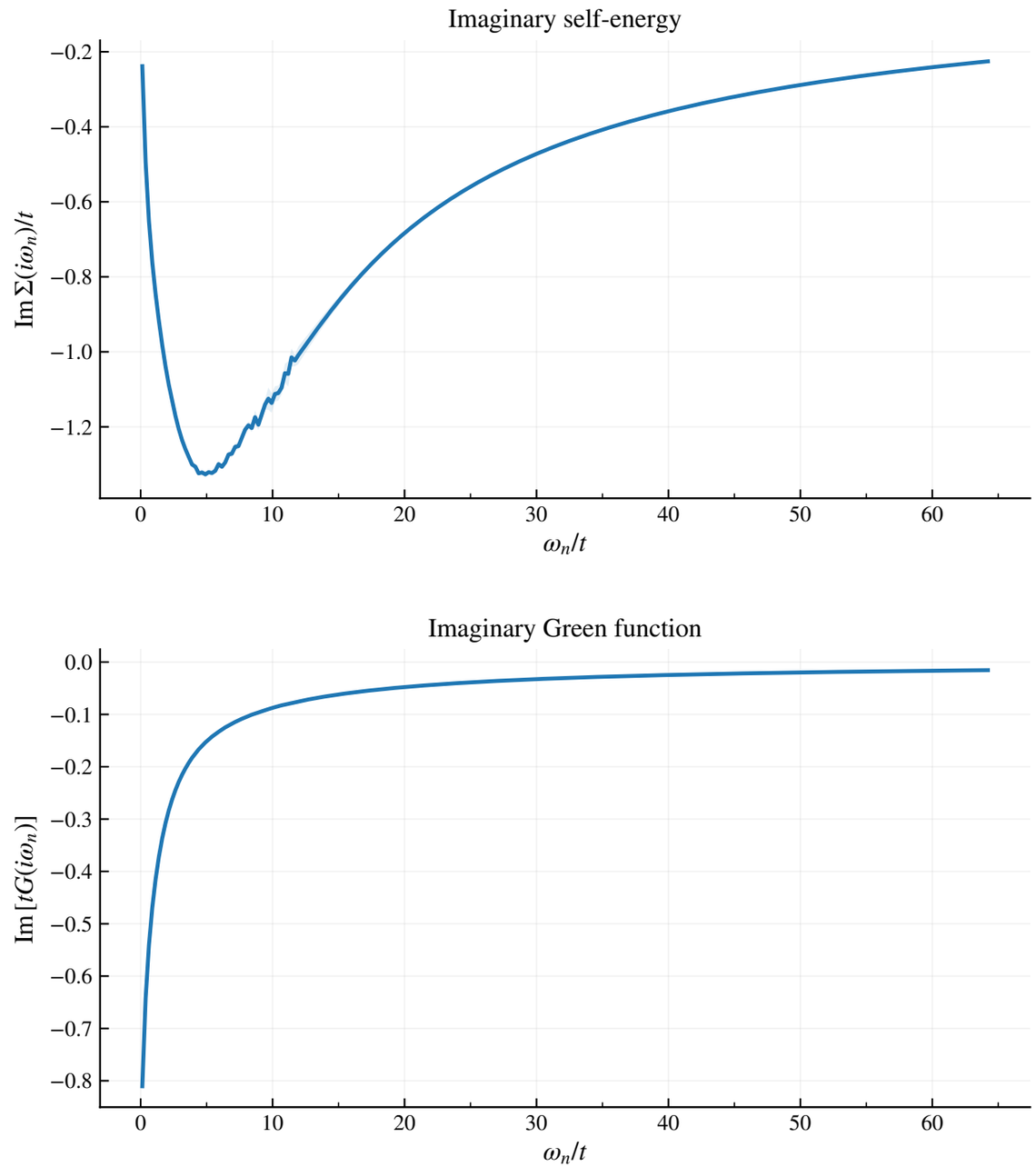
Full positive Matsubara grid | 256 frequencies

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS





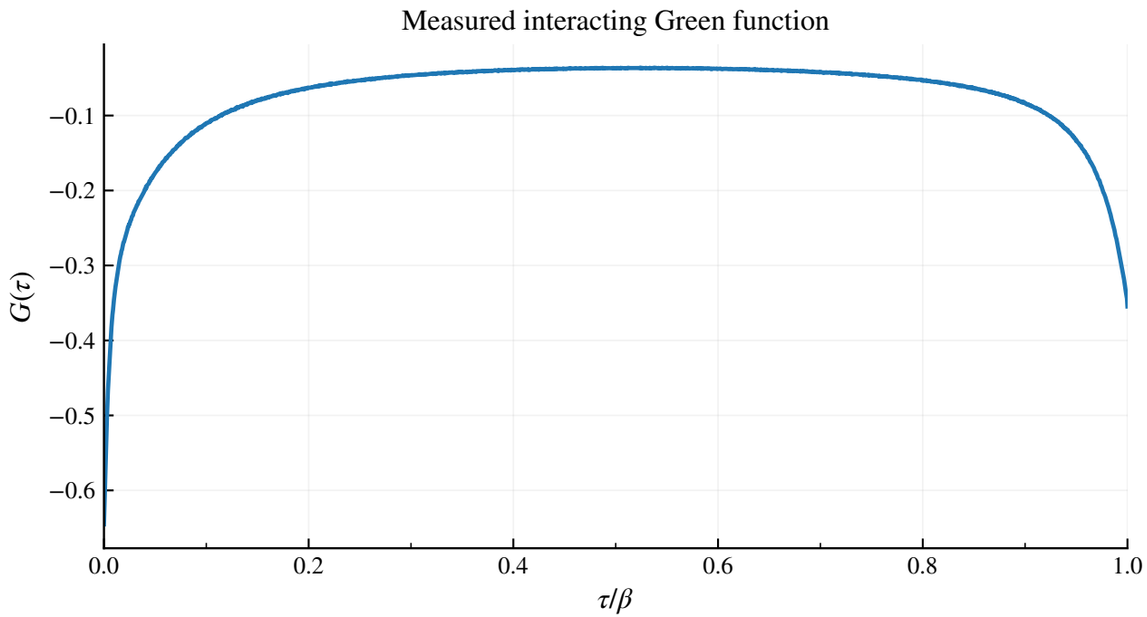
Solid curves: final-replica mean. Shaded envelopes: ±1 replica SEM (when available).

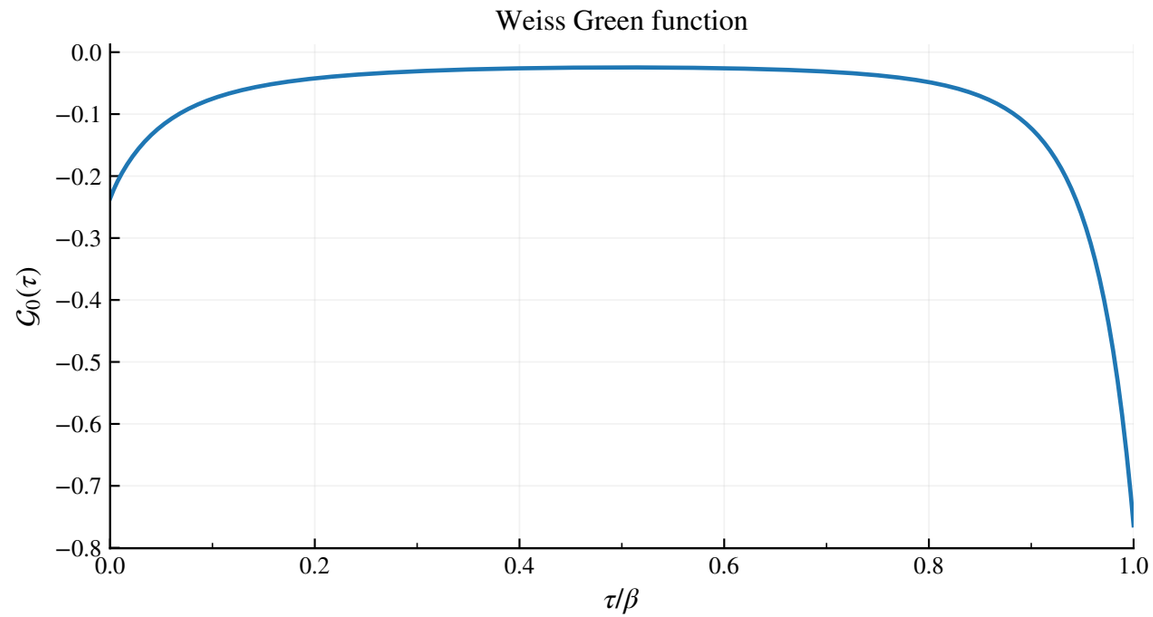
Full imaginary-time grid | 2049 time points

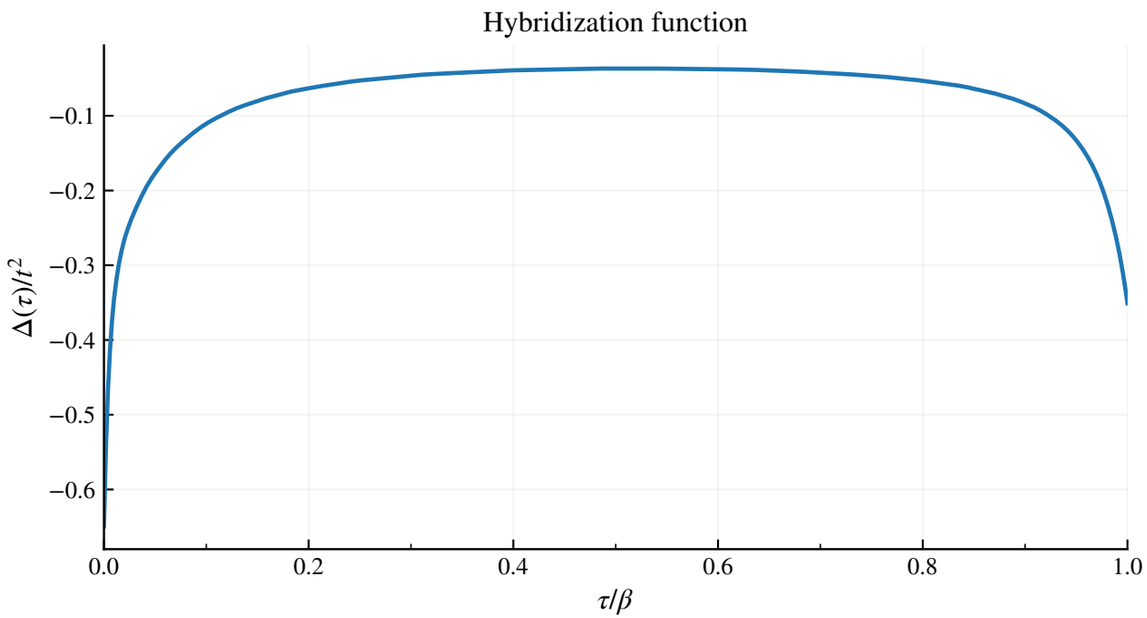
$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

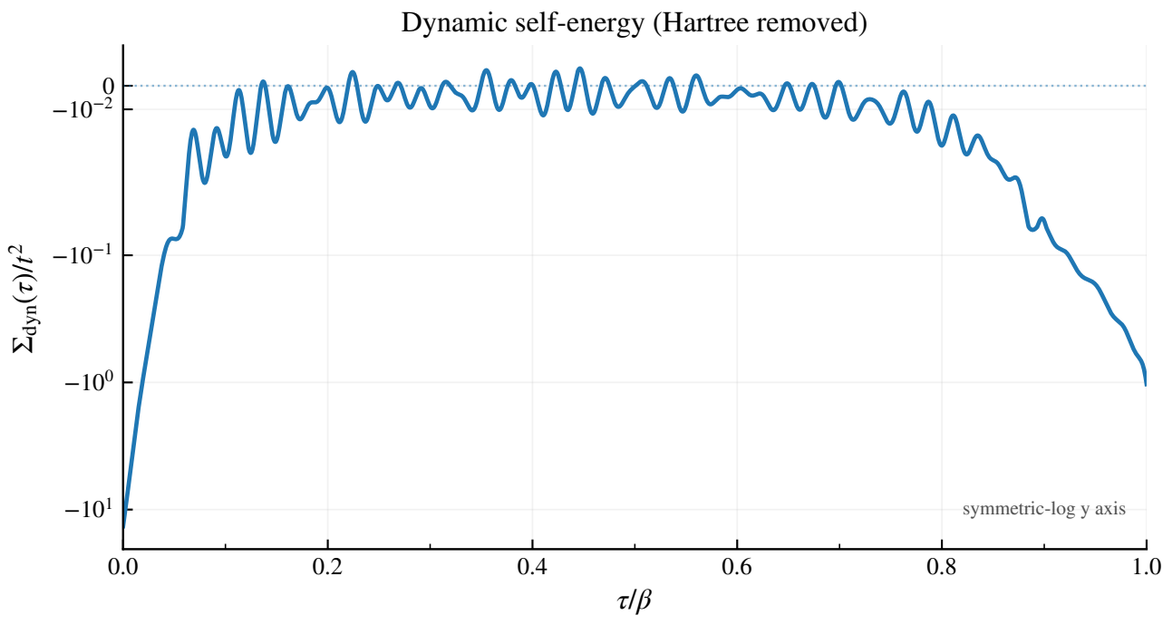
$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS









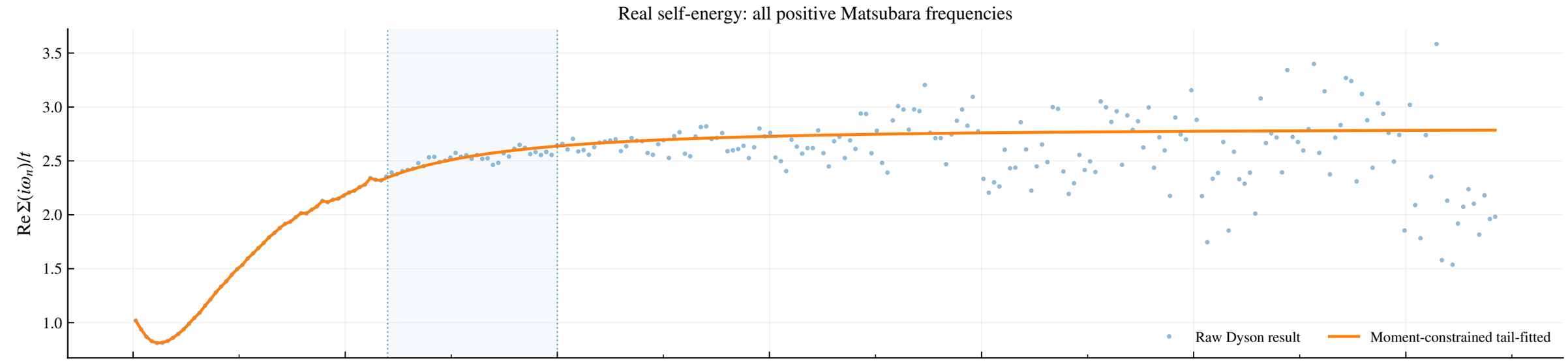
$\Sigma_{\mathrm{dyn}}(\tau)$ excludes the instantaneous Hartree term; it is not the full self-energy.

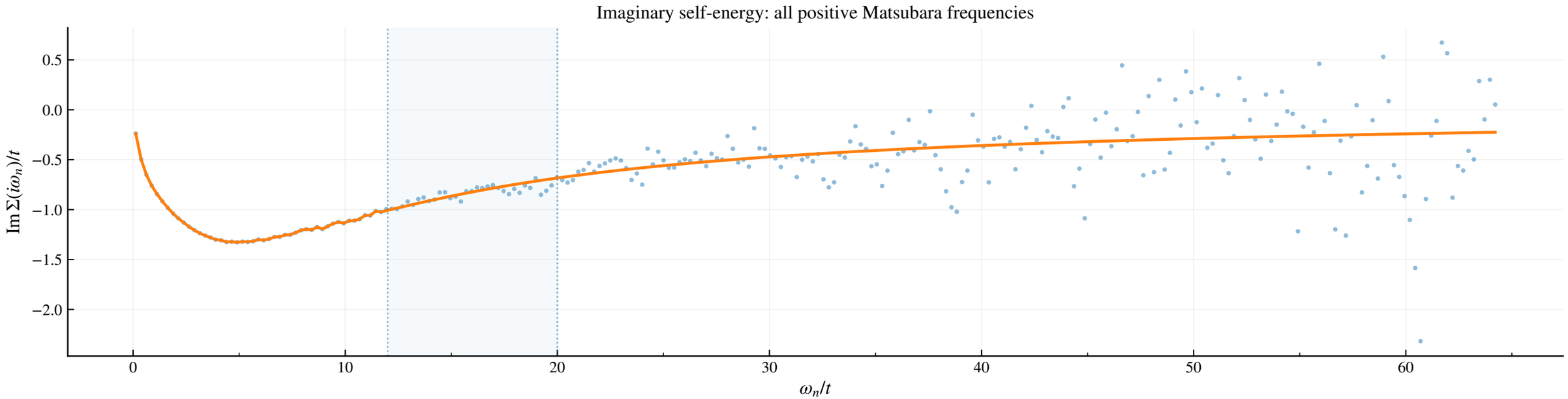
Raw and fitted self-energy | full frequency range

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS





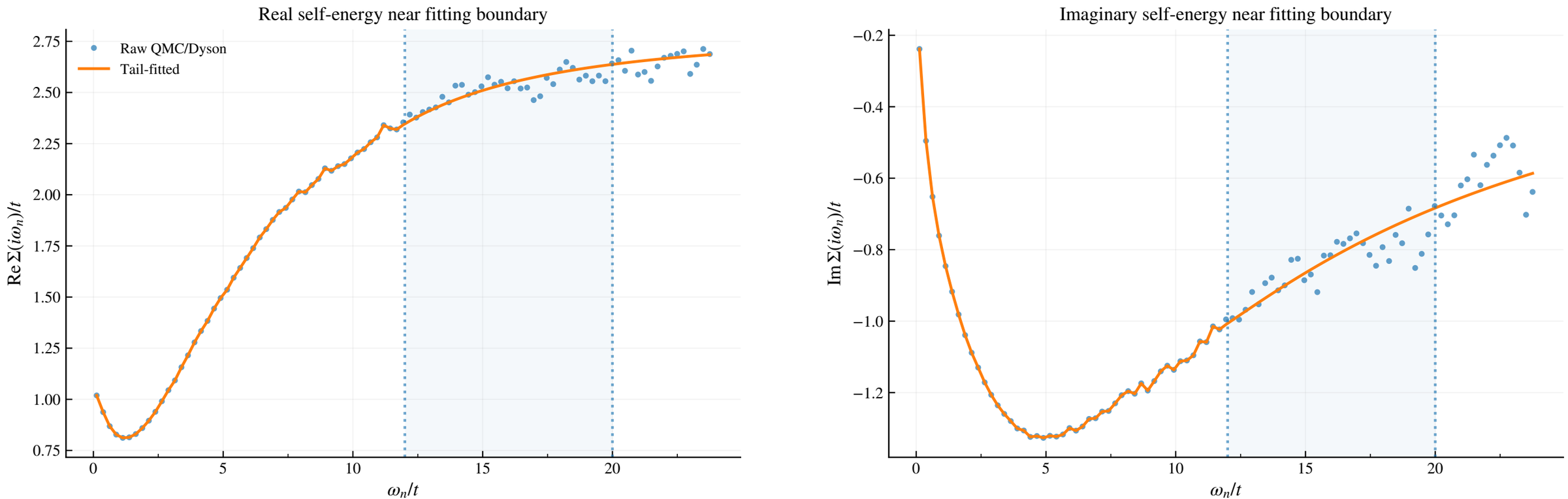
Dotted vertical lines mark the user-selected tail-fit window; raw points are shown without smoothing.

Self-energy fitting region | $0 < \omega_n/t < 24$

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS



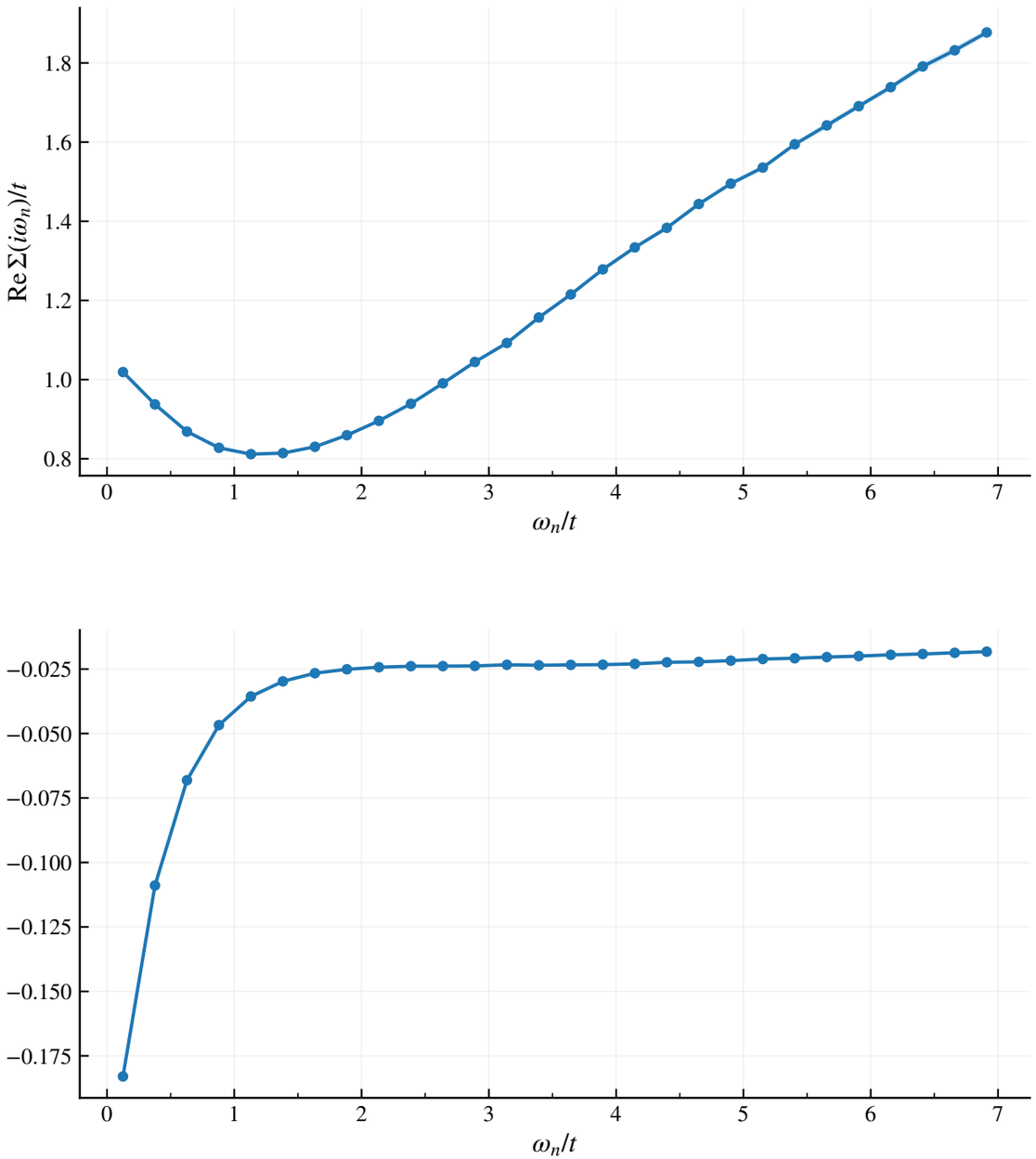
No smoothing of raw data; dotted lines mark the user-chosen tail-fit window.

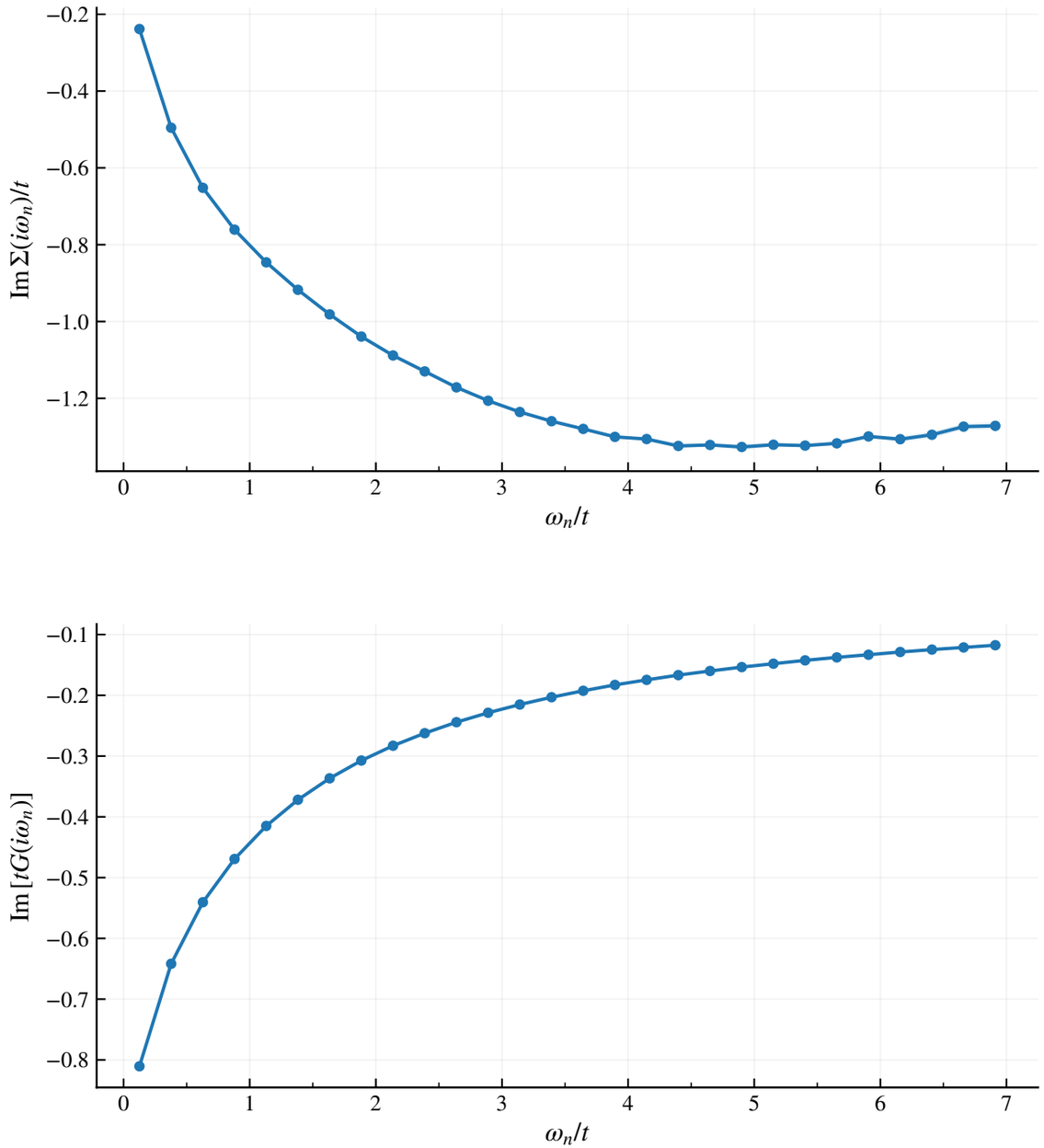
Low-frequency detail | first 28 positive points

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS





Low-frequency quantities are especially relevant for metallicity and quasiparticle diagnostics.

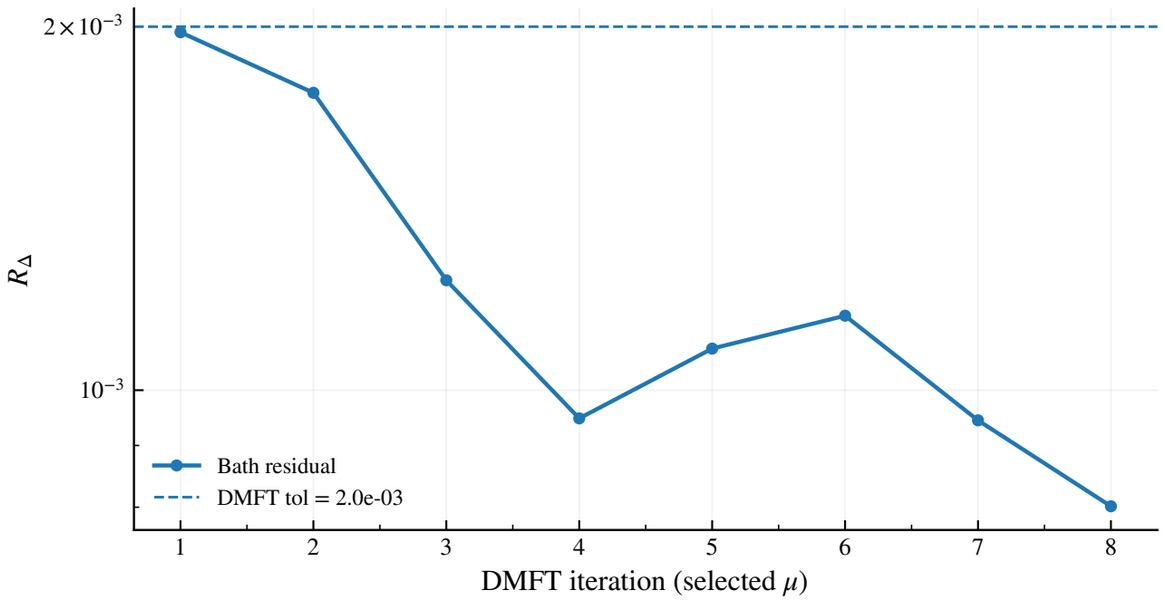
DMFT convergence and sampling diagnostics

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

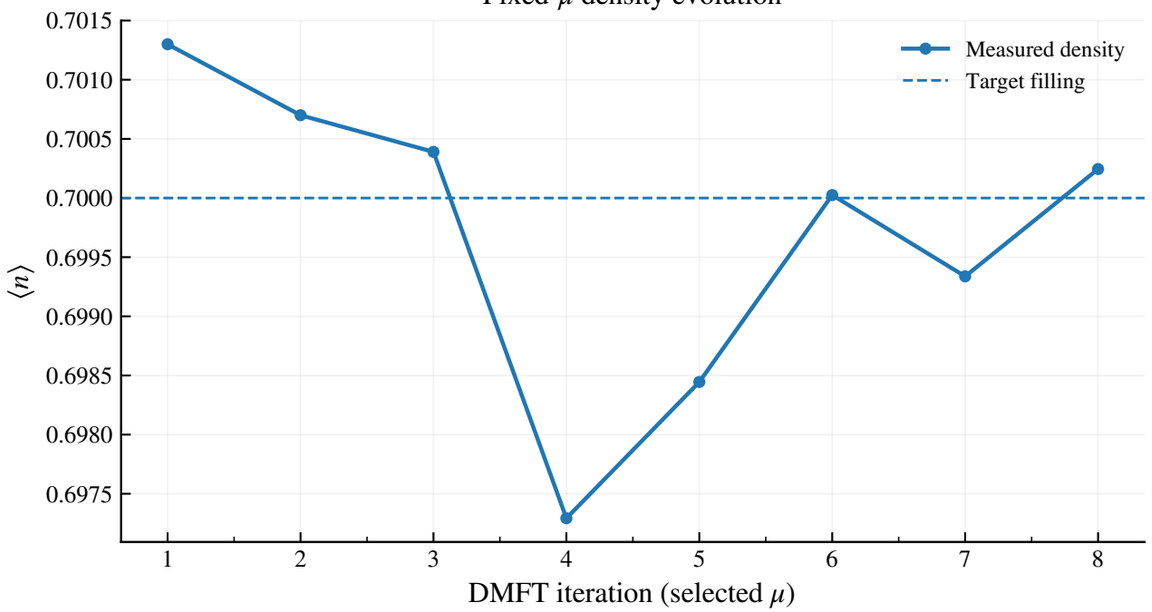
$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS

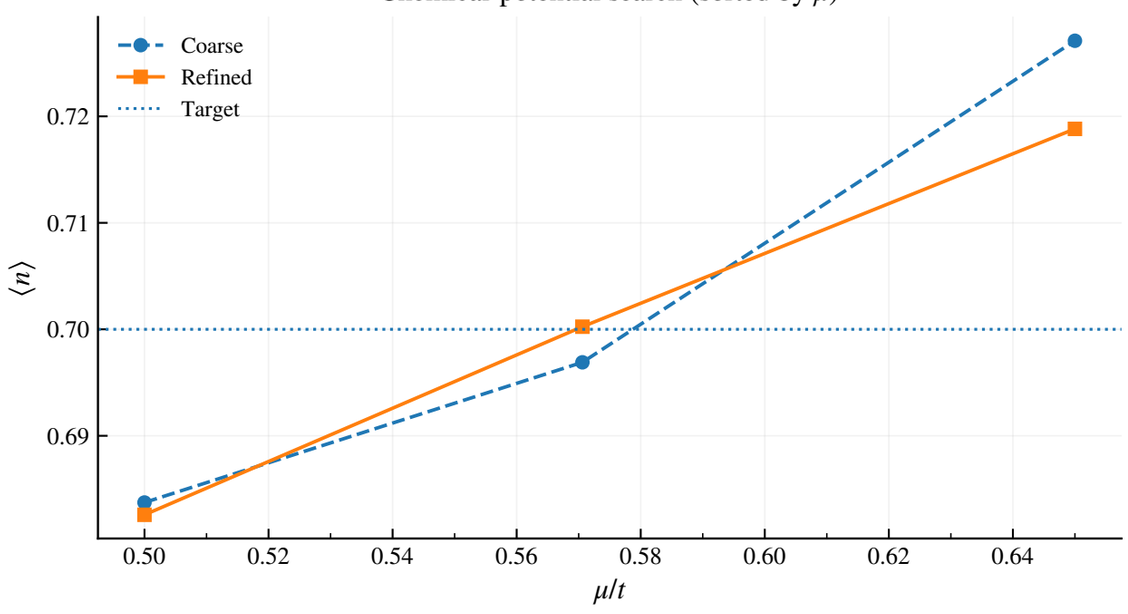
Fixed-μ Bethe convergence



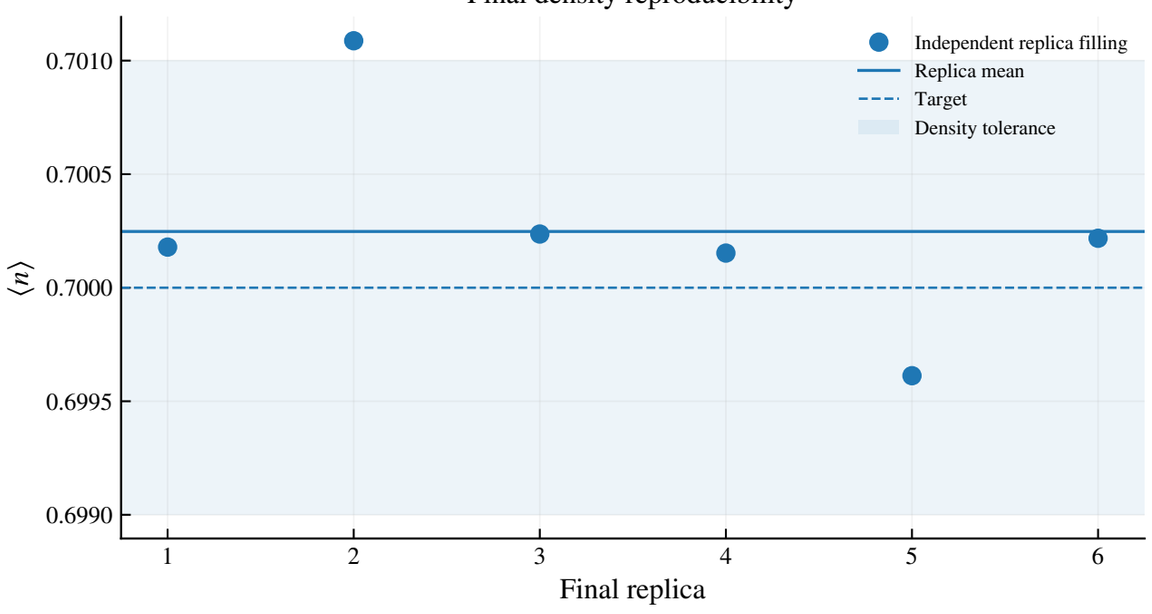
Fixed-μ density evolution



Chemical-potential search (sorted by μ)



Final density reproducibility

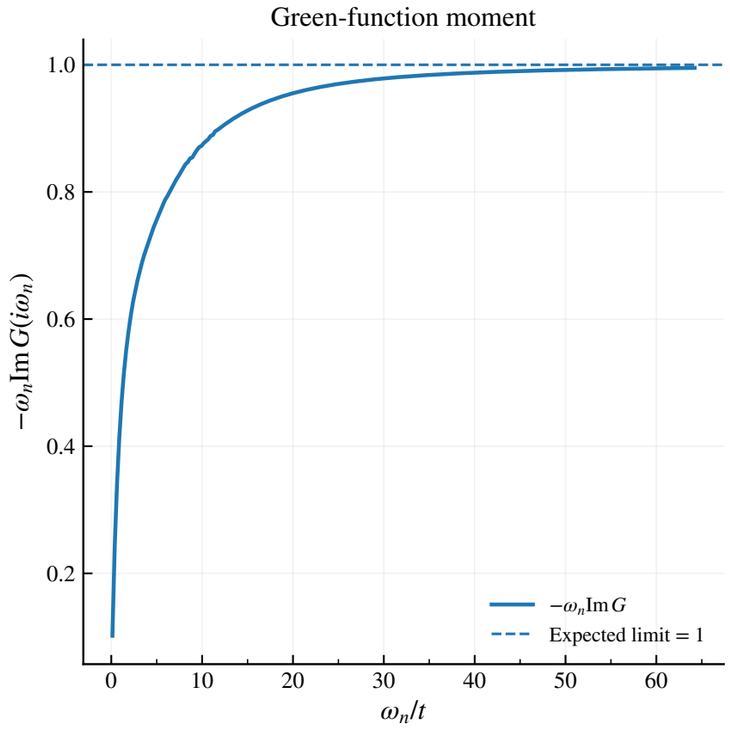


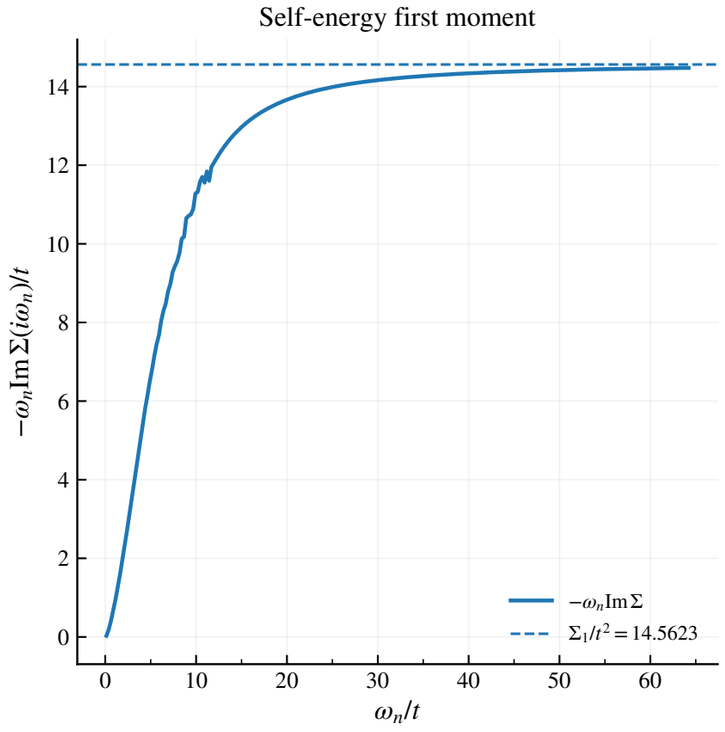
High-frequency consistency checks

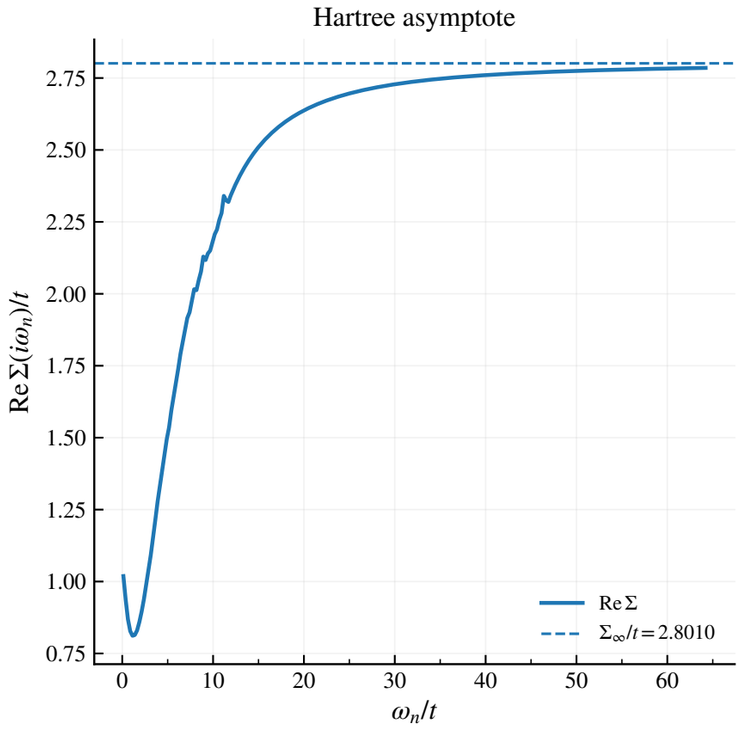
$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS







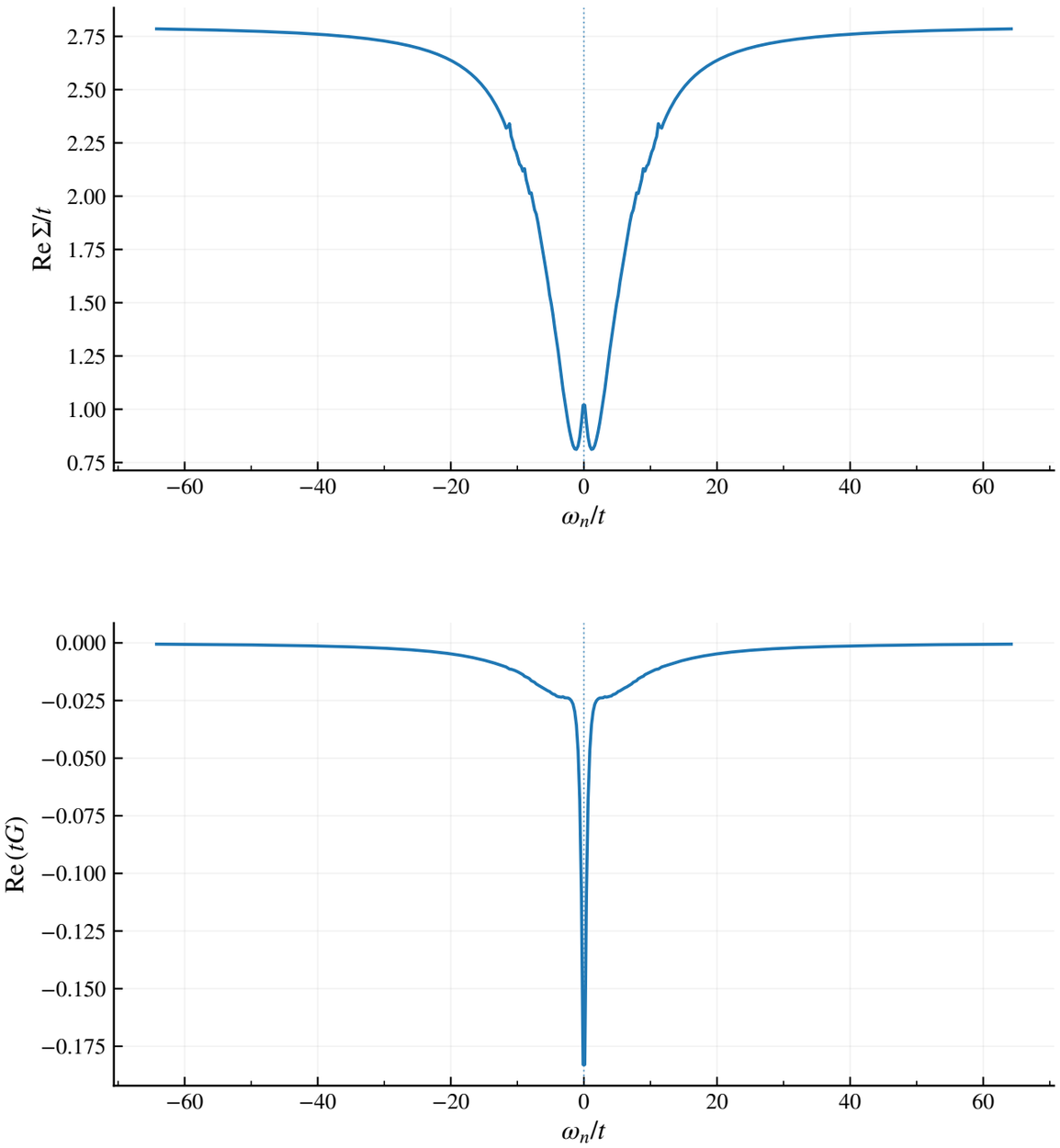
Moment-constrained tail fitting uses known asymptotes; agreement is not an independent measurement of those moments.

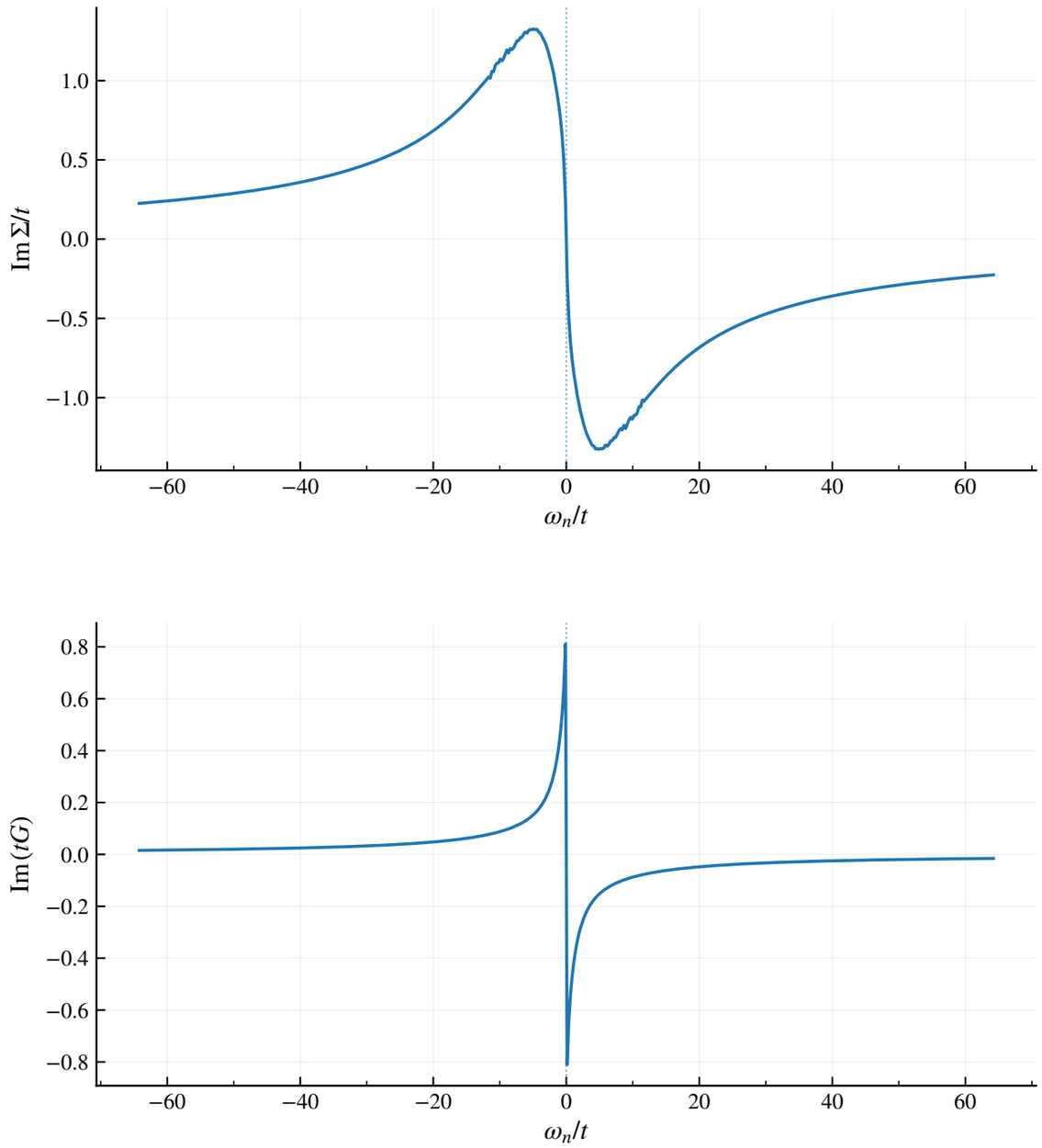
Entire signed Matsubara mesh | 512 frequencies

$U/t = 8.0$ · $\beta t = 25.0$ · $t = 0.25\,\mathrm{eV}$ · $T = 116.05\,\mathrm{K}$

$\langle n \rangle = 0.700248$ (target 0.7000) · $\mu/t = 0.570600$ · $D = 0.00738$ · $R_\Delta = 0.000377$

$\langle \mathrm{sign} \rangle = 1.000$ · $\tau_{\mathrm{auto}} = 0.93$ cycles · 6 final replicas · Quality: PASS





Shows both negative and positive fermionic Matsubara frequencies saved in the NPZ.